RETISS · TEXTILECYCLE

Module M5
Administration, Logistique and the AI voice assistant

What was built, how it works, and how to test every feature and option, step by step.

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Project	RETISS / TextileCycle — circular-economy platform for textiles (Laravel 12), ESPRIT 5TWIN1, team CodeCraft
Scope	Shared User module, part M5: admin dashboard (user management and statistics). Individual module M5: Logistique (Tournee ↔ Mission), with automatic assignment, real-time tracking, map, route optimisation and donation-volume forecasting. Extra: AI voice assistant for both spaces.
Date	02 November 2026

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1. Overview

The project PDF splits the work in two. Every member owns one part of the shared **User** module, and one individual business module. As **M5**, my parts are:

Part	What the project PDF asks for	What was delivered
User module — M5	Admin dashboard: user management, statistics	Full user CRUD with search, filters and pagination; statistics page with charts; custom admin interface (action bar, confirmation popups, custom dropdowns).
Individual — M5 Logistique	Entities Tournee ↔ Mission (CRUD, 1-N); advanced: automatic assignment, real-time tracking, map; AI: route optimisation and donation-volume forecasting	Tournee/Mission CRUD linked to real orders (M2) and donations (M1); automatic assignment; phone "tour mode" with GPS; live admin map; route optimisation with an AI explanation; donation forecast with an AI briefing; printable route sheet.
Extra	—	AI voice assistant (French and English) that drives both spaces and answers questions about the data.

1.1 Where the code is

- **GitHub, branch** `Imem-Ayachi-M5` (pushed): commit `a521a14` — admin dashboard and Logistique CRUD; commit `24c2168` — AI voice assistant. Both are already merged into the team branches `ala` and `integration`.
- **Local branch** `Imem-Ayachi-M5-avance` (created from `integration`): all the advanced Logistique features (sections 5 to 7 of this document). At the time of writing this work is **not yet committed**.

1.2 Technical choices in one paragraph

Everything runs on the existing stack (Laravel 12, Blade, Tailwind from the CDN, SQLite) with **no paid service**. Maps use Leaflet with OpenStreetMap tiles; addresses are geocoded with OpenStreetMap Nominatim. The "AI" is of two kinds, deliberately: **algorithms** where results must be exact and verifiable (route optimisation, assignment, forecasting), and a **language model** (free Groq API) where words are needed (voice commands, explanations, briefings). The language model never computes a number shown to the user: it only explains numbers computed by the code, and every action it proposes is re-validated by the server and confirmed by the user before anything changes.

2. Installation and demo data

2.1 Requirements

- PHP 8.2+ with the extensions `pdo_sqlite`, `gd`, `zip` and `intl`. With XAMPP, open `C:\xampp\php\php.ini` and remove the ";" in front of `extension=gd`, `extension=zip` and `extension=intl` (the team's QR-code and Excel packages need them).
- Composer, and Chrome or Edge (voice dictation and the map work best there).
- Optional, for AI features: a free Groq API key from console.groq.com/keys. Without it, every page still works; AI texts are replaced by computed summaries and the assistant says it is not configured.

2.2 Commands

```
composer install
copy .env.example .env          (then edit .env, see below)
php artisan key:generate
php artisan migrate:fresh --seed
php artisan serve                (open http://127.0.0.1:8000)
```

Settings added to `.env` by my module (all have working defaults):

Variable	Purpose
ASSISTANT_API_KEY	Groq key for the voice assistant and the AI explanations. Empty = AI off.
ASSISTANT_MODEL	Language model (default `openai/gpt-oss-120b`).
ASSISTANT_TRANSCRIPTION_MODEL	Speech-to-text model (default `whisper-large-v3-turbo`).
ASSISTANT_VOICE_MODEL / ASSISTANT_VOICE	Natural English voice (Orpheus, voice "hannah"). Empty model = browser voice only.
GEOCODAGE_ACTIF	Turns address geocoding (OpenStreetMap) on or off.
LOGISTIQUE_DEPOT_NOM / _LAT / _LNG	Depot where tournées start and end (default: RETISS warehouse, Charguia).

Note. the natural English voice (Orpheus) only works after the owner of the Groq account accepts the model's terms once, at console.groq.com/playground?model=canopylabs%2Forpheus-v1-english. Until then the assistant uses the browser voice.

2.3 Demo accounts

Account	Password	Use it for
admin@retiss.tn	password	Everything under /admin: users, statistics, logistics, live map, assignment, forecast.
collecteur@retiss.tn	password	Collecteur with today's tournée "Tunis Centre — Lac" (in progress, 5 stops).
nour.collecte@retiss.tn	Password123!	Collecteur with today's "Le Bardo" and tomorrow's "Grand Tunis" (the one to optimise).
...@demo.retiss.tn	Password123!	Demo donors and clients created by my seeder (e.g. rania.toumi@demo.retiss.tn).

The team seeder `TestAccountsSeeder` runs after mine and resets the admin and `collecteur@retiss.tn` passwords to `password`; that is why two passwords exist.

2.4 What the demo data contains

My seeders (`LogistiqueSeeder` and `HistoriqueDonsSeeder`) create realistic data on Greater Tunis, with real coordinates. Dates are relative to the day you run the seeders:

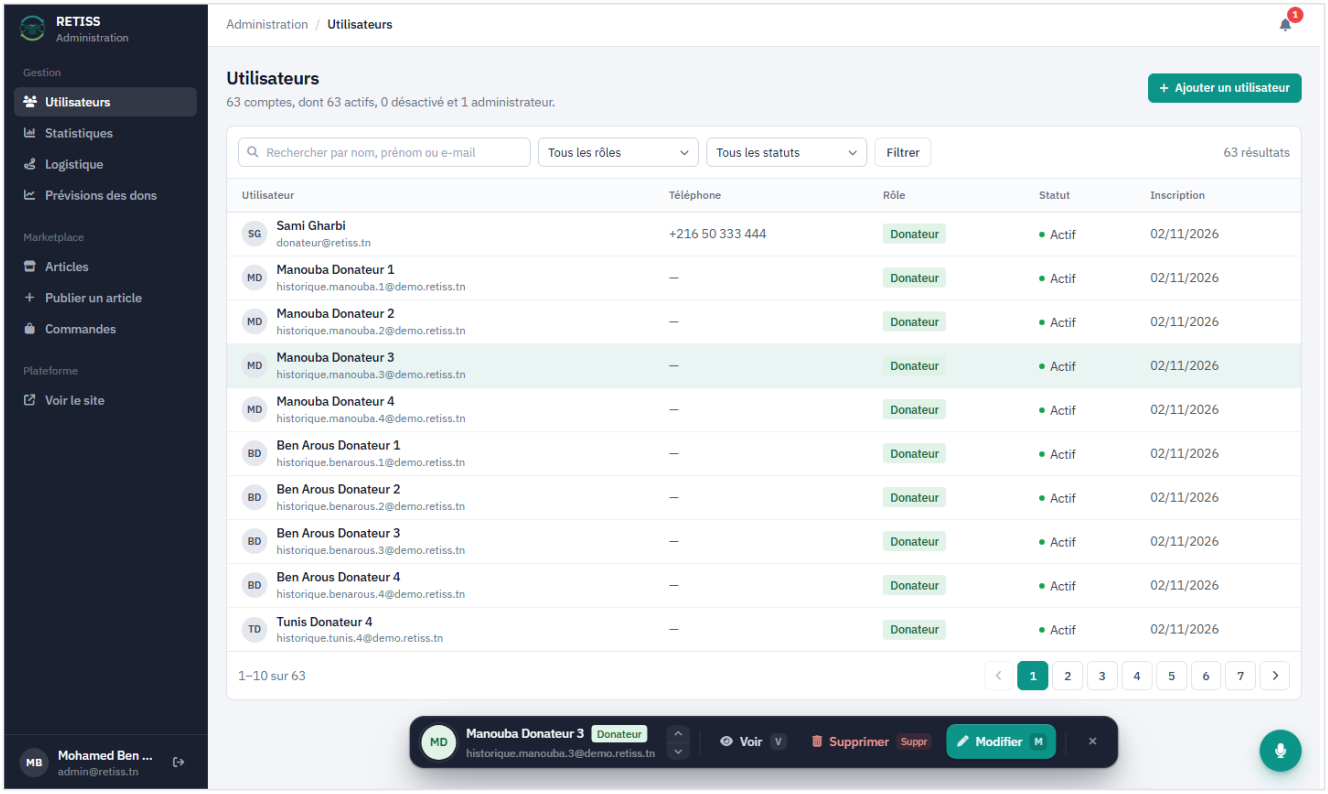
Day	Tournée	Collecteur	Content
Yesterday	Ariana — Ennasr	collecteur@	Completed: 2 pickups done, 1 delivery failed ("Client absent").
Today	Tunis Centre — Lac	collecteur@	In progress: 2 done, 1 in progress, 2 to do. Live GPS position near Montplaisir.
Today	Le Bardo	nour.collecte@	1 pickup done, 1 delivery to do. Last GPS position 12 minutes old.
Tomorrow	Grand Tunis	nour.collecte@	6 deliveries entered in a zigzag order (128.9 km): the optimisation demo.
In 2 days	La Marsa — Carthage	collecteur@	1 pickup, 1 delivery, planned.

- **Waiting to be planned**: 5 paid orders and 1 donation that are in no tournée (used by automatic assignment).
- **Donation history**: about 1,250 past donations over 66 weeks in four governorates, with growth and seasons (used by the forecast).
- Orders, donations and stops are linked: finishing a seeded delivery already marked its order as delivered, etc.

3. Admin dashboard (User module, part M5)

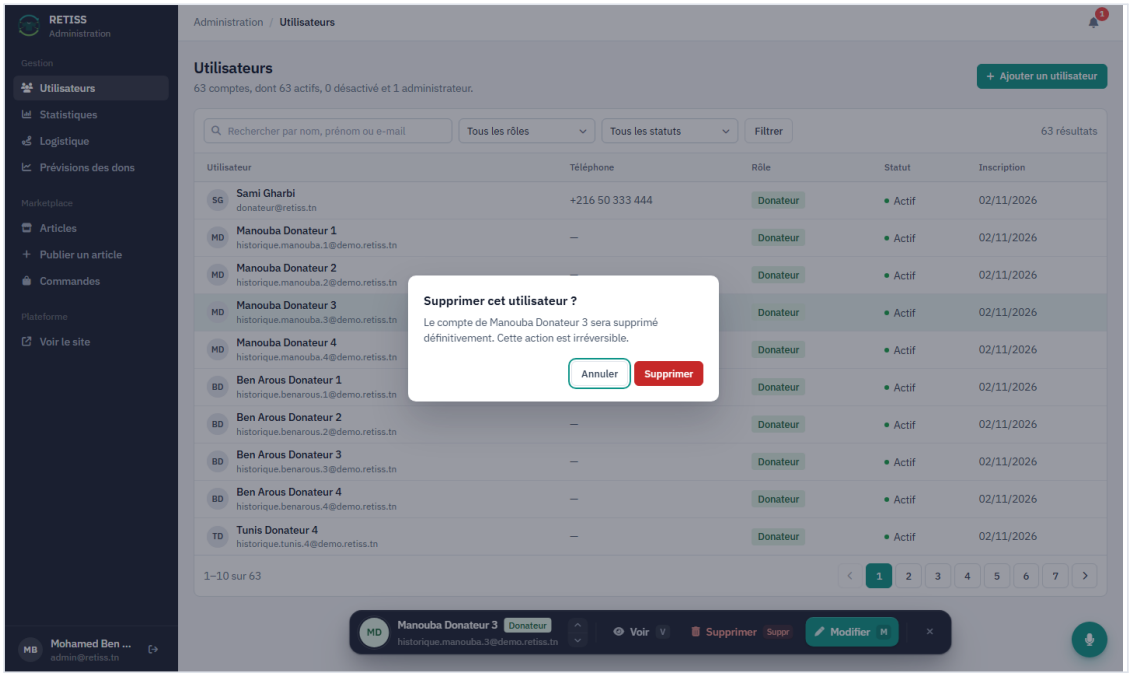
Reserved to the **ADMIN** role (middleware `EnsureAdmin`: any other role gets a 403 page). It has its own interface: a dark sidebar, a slim top bar with the breadcrumb and the team's notification bell, and a small design system shared by all my pages.

3.1 User management (CRUD)



Users list. Clicking a row slides up the action bar (bottom); role badges are colour-coded.

- **List** (`/admin/users`): search by first name, last name or email; filter by role and by status (custom dropdowns, the filter applies as soon as you pick); 10 users per page; a summary line with the counts.
- **Action bar**: click any row → a bar slides up from the bottom with the user, their role, and the actions Voir (V), Modifier (M), Supprimer (Suppr). Arrow keys or the small up/down stepper move to the previous/next user; Escape closes it.
- **Create / edit**: first name, last name, email, phone, role, active account, password (required on create, optional on edit: empty keeps the current one). Validation messages are in French and shown under each field.
- **Delete**: always asks through a custom confirmation popup (no browser alert).
- **Safety rules**: an admin cannot delete their own account, demote themselves, or deactivate themselves.

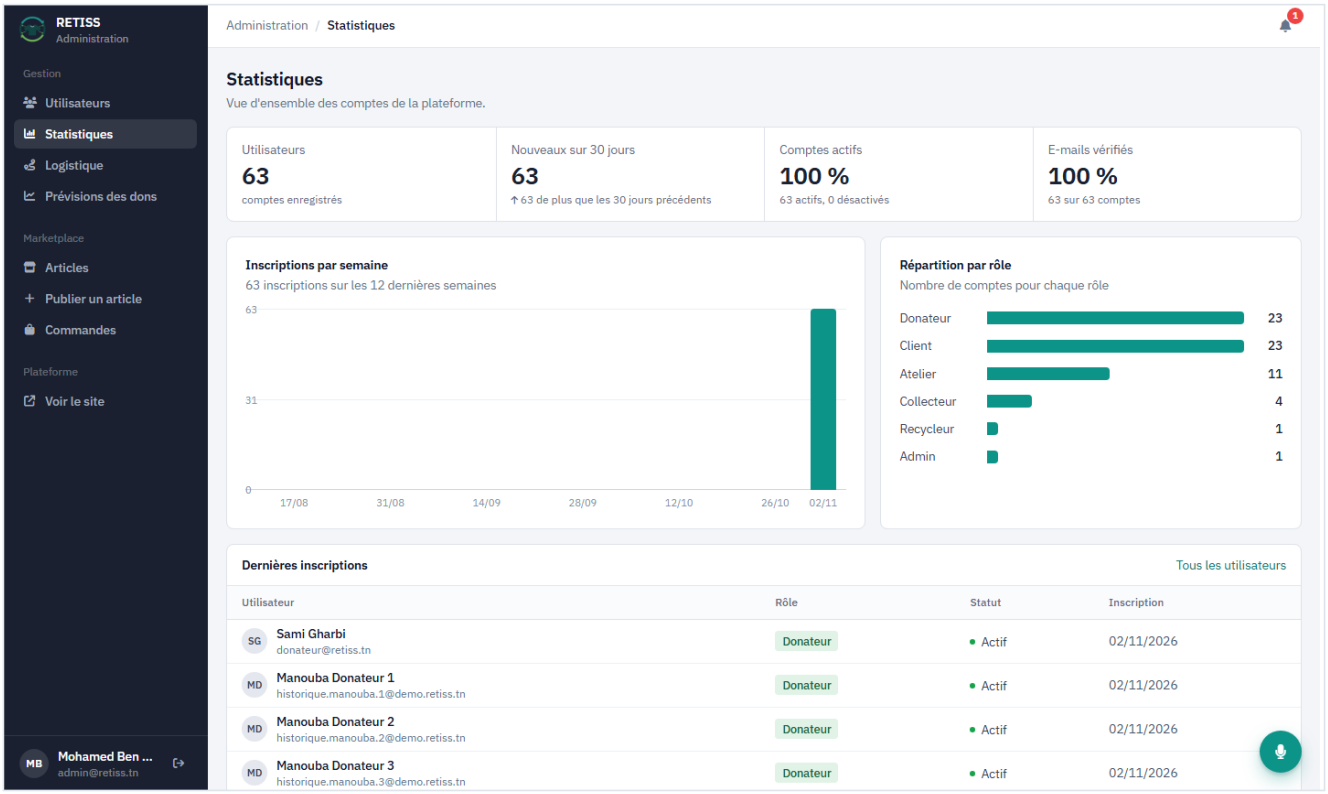


Deleting always goes through the custom confirmation popup.

How to test

#	Do this	You should see
1	Log in as admin@retiss.tn, click **Utilisateurs** in the sidebar.	The list, with the count summary at the top and pagination at the bottom.
2	Type part of a name in the search box and press Enter. Then pick a role in "Tous les rôles".	Only matching users remain; the filter applies immediately on pick.
3	Click **Ajouter un utilisateur**, submit the empty form.	Red messages under each required field (Prénom, Nom, E-mail, Mot de passe).
4	Fill it with an email that already exists.	"Cette adresse e-mail est déjà utilisée."
5	Create a valid user (role Collecteur, password + confirmation).	Redirect to the user's page with a green success message.
6	Back in the list, click a row, then press **M** (or click Modifier). Change the role, leave the password empty, save.	Role updated; the old password still works.
7	Click a row, then **Supprimer**; click Annuler, then repeat and confirm.	Nothing deleted on Annuler; the user disappears after confirming.
8	Click your own row (marked "vous").	The action bar has no Supprimer button.
9	Edit your own account and set the role to Client or untick "Compte actif".	Red message: you cannot change your own role or deactivate your account.
10	Log in as a collecteur and open http://127.0.0.1:8000/admin/users.	403 page.

3.2 Statistics



Statistics: key figures, sign-ups per week, accounts per role, latest sign-ups.

- Four key figures: total accounts, new accounts over 30 days compared with the 30 days before, active share, verified emails.
- Two charts drawn without any chart library: sign-ups per week over 12 weeks, and accounts per role (largest first). Hovering or focusing a bar shows a tooltip.
- The six latest sign-ups, each linking to the user page. The same numbers feed the voice assistant (section 6).

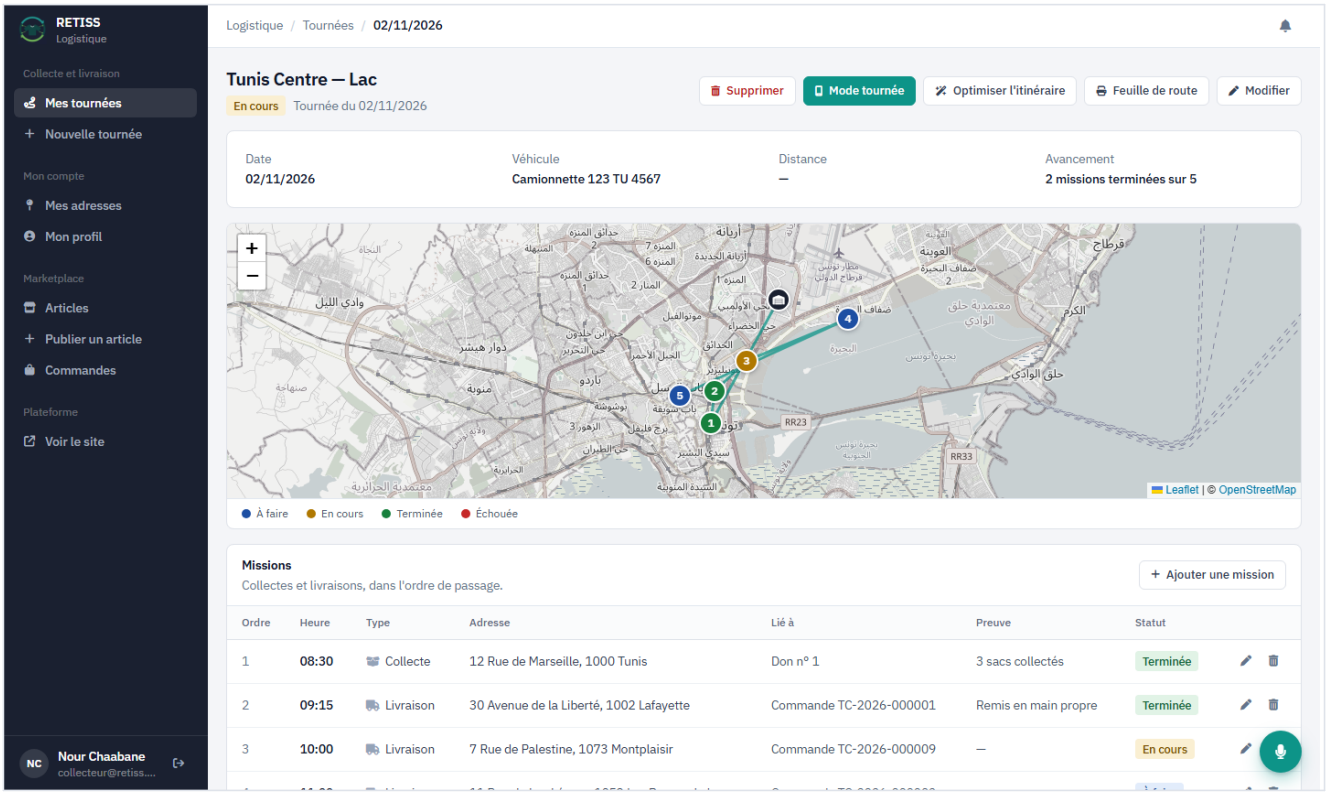
How to test

#	Do this	You should see
1	Click **Statistiques**.	The four figures and both charts.
2	Hover a weekly bar and a role bar.	A dark tooltip with the week (or role) and the exact count.
3	Create a user (3.1), come back.	Total and "Nouveaux sur 30 jours" increased by one; the user is first in "Dernières inscriptions".

4. Logistique: tournées and missions (CRUD)

Reserved to the **COLLECTEUR** role (middleware `EnsureCollecteur`). A collecteur only ever sees and edits their own tournées: opening someone else's gives a 403. A **tournée** is one day's round in one zone; a **mission** is one stop of that round, either a pickup (Collecte) or a delivery (Livraison). One tournée has many missions.

Entity	Main fields
Tournee	collecteur, date, zone, vehicle, distance (km), status (Planifiée, En cours, Terminée, Annulée), itinéraire IA (summary of the last optimisation)
Mission	tournée, type (Collecte / Livraison), address, GPS position and its source, stop number, planned time, status (À faire, En cours, Terminée, Échouée), proof note and photo, finish time, link to an order (M2) or a donation (M1)



A tournée: status, information, map of the stops coloured by status, and the stops with what they are linked to.

How to test

#	Do this	You should see
1	Log in as collecteur@retiss.tn (password: password) and click **Mes tournées**.	The collecteur's tournées, with a search box (zone or vehicle) and a status filter.
2	Click **Nouvelle tournée**, submit empty, then fill date, zone, vehicle.	Field errors first; then the new tournée page with an empty stops list.
3	On the tournée, click **Ajouter une mission**.	The mission form; the stop number is pre-filled with the next one.
4	Choose type Livraison, pick an order in "Commande à livrer", leave the address empty, add a time, save.	The address of the order is used automatically; the stop appears in the list and on the map.
5	Edit the mission and change its status.	The badge changes; the tournée status follows (see 5.1).
6	Delete a mission, then delete a tournée.	Custom confirmation popup each time; deleting a tournée deletes its missions.
7	Open http://127.0.0.1:8000/logistique/tournees/4 (nour.collecte's tournée) as collecteur@retiss.tn.	403 page.

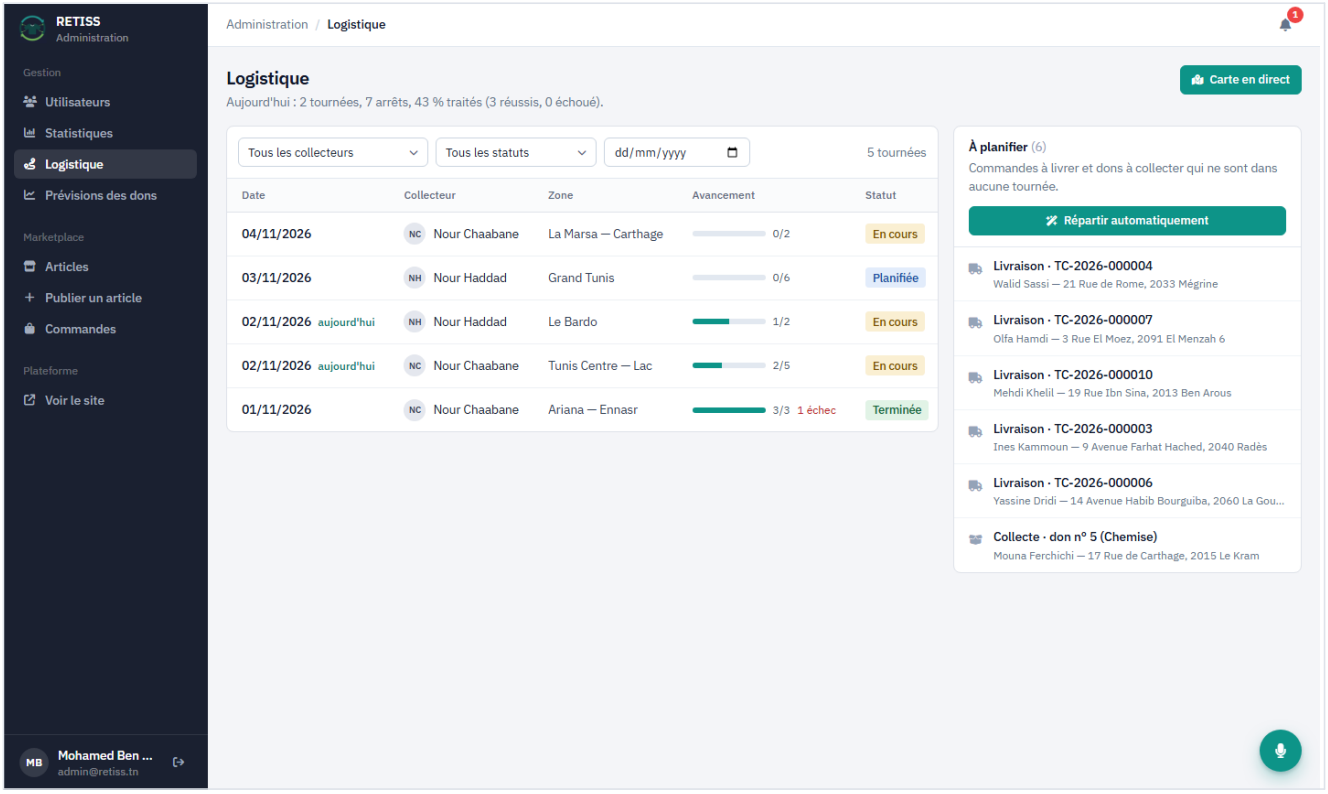
5. Advanced Logistique features

5.1 Real data: locations and synchronisation

Each stop gets a GPS position automatically, from the most reliable source available: the linked order's delivery address, the donor's default address for a pickup, or the typed address geocoded with OpenStreetMap (results cached, at most one request per second as its usage policy requires). A stop that cannot be placed is flagged "Introuvable sur la carte". When a stop changes status, the linked data follows, wherever the change comes from (form, phone tour mode or voice assistant):

Stop event	What happens automatically
Delivery started	Order → EXPEDIEE, entry in its history, client notified ("your order is on its way").
Delivery done	Order → LIVREE, proof note copied into its history, client notified.
Delivery failed	Reason written in the order history, client notified, order goes back to "À planifier", admins alerted.
Pickup done	Donation DEPOSE → EN_TRI (arrives at sorting).
First stop started / last stop closed	Tournée → En cours / Terminée.

Validation also prevents mistakes: an order can only go on a delivery and a donation only on a pickup, and the same order or donation cannot be in two active missions (after a failure, a new attempt is allowed).



Admin logistics overview: today's progress, every collecteur's tournées, and the jobs nobody has planned yet.

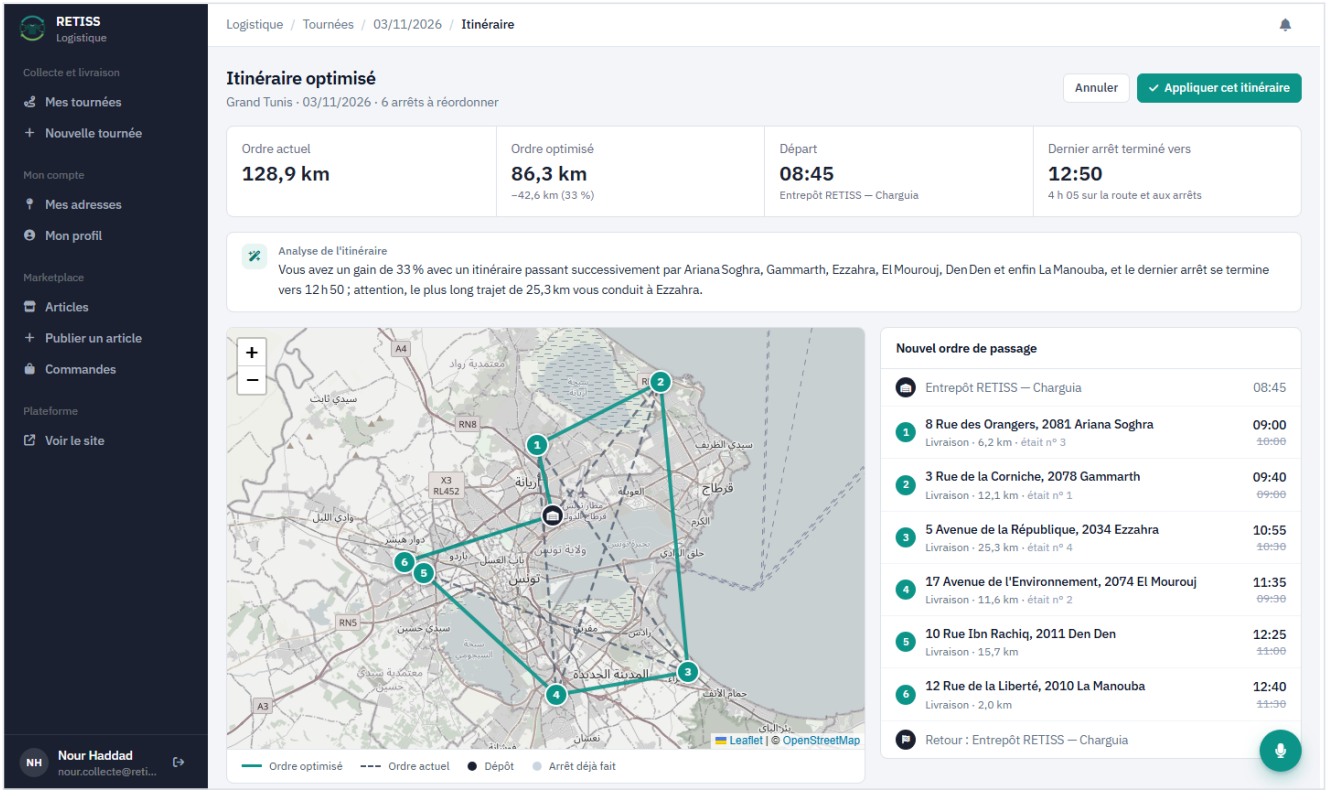
How to test

#	Do this	You should see
1	As admin, click **Logistique** in the sidebar.	Today's summary, all tournées with a progress bar, and "À planifier (6)" on the right.
2	As collecteur@retiss.tn, open "Tunis Centre — Lac" and edit stop 3 (the one "En cours"): set it to Terminée.	The stop is Terminée. Log in as nadia.zouari@demo.retiss.tn (Password123!) and open Commandes: the order is "Livrée", with the note in its history.
3	Add a mission of type Collecte and try to pick an order.	The order field is hidden for a pickup; a donation field is shown instead.
4	Try to put an order that is already in an active mission on a new mission.	"Cette commande est déjà prise en charge par une autre mission."
5	Mark a delivery as Échouée with a reason.	As admin: the bell shows a new alert; the order is back in "À planifier".

5.2 Route optimisation (AI)

Algorithm. Finding the shortest order of stops is the travelling-salesman problem. The optimiser first builds a route by always going to the nearest remaining stop (nearest neighbour), then improves it with **2-opt**: it reverses parts of the route as long as that makes it shorter, which removes every crossing. Distances use the haversine formula × 1.3 to approximate real roads; times assume 25 km/h in town plus 10 minutes per stop, rounded up to 5 minutes. Finished stops keep their place, the stop in progress stays first, and the route restarts from the last finished stop (or the depot). Stops without a position go last.

AI explanation. The language model receives only computed facts (gain, order of neighbourhoods, longest leg, finish time) and writes a short explanation; it is forbidden to invent geography or traffic.



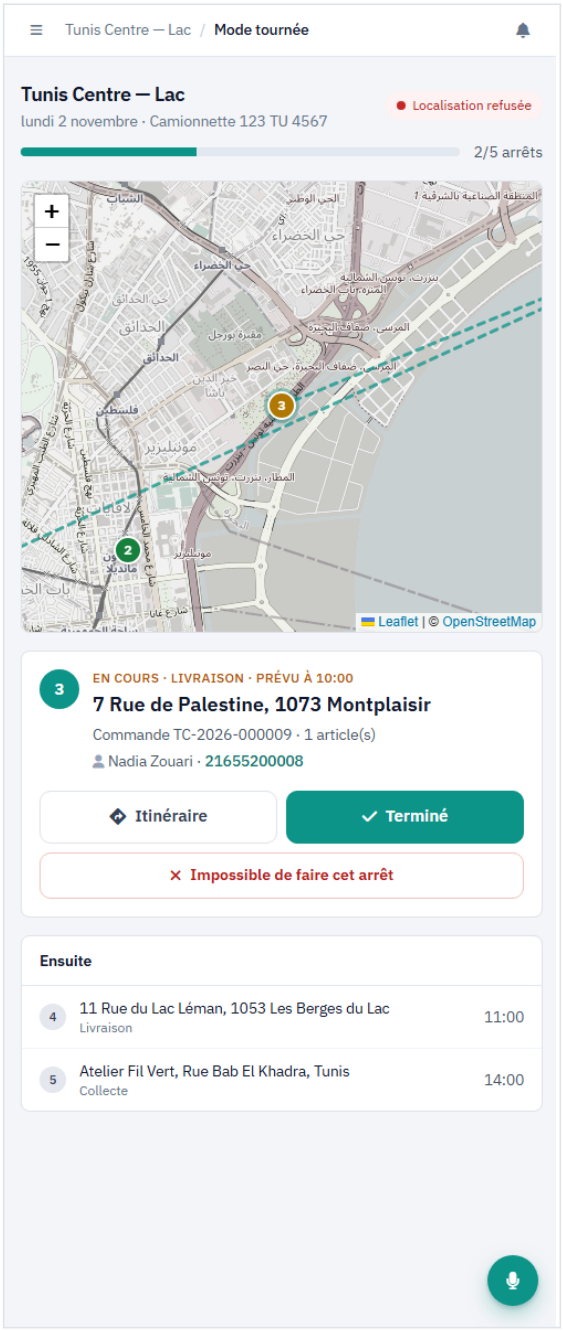
Preview: current vs optimised distance (128.9 km → 86.3 km, −33 %), AI explanation, map (dashed = current order), new order and times.

How to test

#	Do this	You should see
1	Log in as nour.collecte@retiss.tn (Password123!), open tomorrow's tournée "Grand Tunis".	Six deliveries in a zigzag order; an **Optimiser l'itinéraire** button.
2	Click **Optimiser l'itinéraire**.	Preview: 128.9 km → 86.3 km (−33 %), explanation, map, new order with old times struck through. Nothing has changed yet.
3	Click **Annuler**, then reopen the tournée.	The original order is untouched.
4	Open the preview again in two tabs; in one tab, edit a stop; in the other, click **Appliquer**.	"La tournée a changé entre-temps : voici la proposition mise à jour." — an outdated plan is never applied.
5	Click **Appliquer cet itinéraire**.	Green message "42.6 km économisés"; stops renumbered, times updated, distance filled.
6	Optimise today's "Tunis Centre — Lac" as collecteur@retiss.tn.	The stop in progress stays first; the route starts from the last finished stop, at the current time.

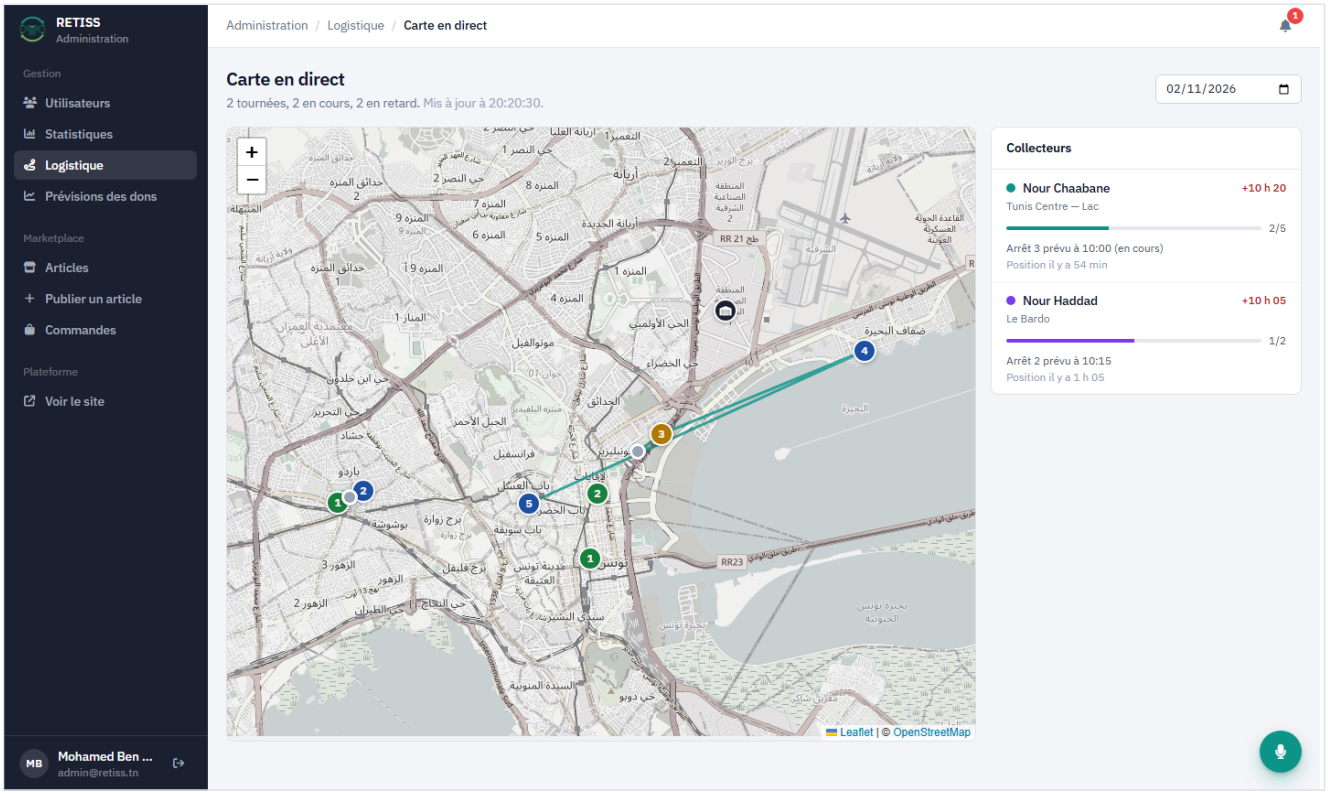
5.3 Tour mode (phone) and real-time tracking

Mode tournée is a phone-first screen: one stop at a time with its address, what it carries, the contact and a tap-to-call number, a **Itinéraire** button that opens the phone's maps app, and large buttons: Démarrer, Terminé (optional note and photo taken with the camera) and "Impossible de faire cet arrêt" (reason required, with quick choices). While the screen is open on the day of the tournée, the phone sends its position about every 20 seconds (the server keeps at most one every 15 seconds).



Tour mode on a phone-sized screen.

Carte en direct (admin) shows every tournée of a day on one map, each with its own colour, plus each collecteur's last position: pulsing blue if under 5 minutes old, grey otherwise. The side list shows progress, the next stop and lateness ("+9 h 07" when the next stop's planned time has passed). It refreshes every 10 seconds while the tab is visible; another date can be picked to replay a day.



Live map: two tournées today, live position (blue), old position (grey), lateness in the side list.

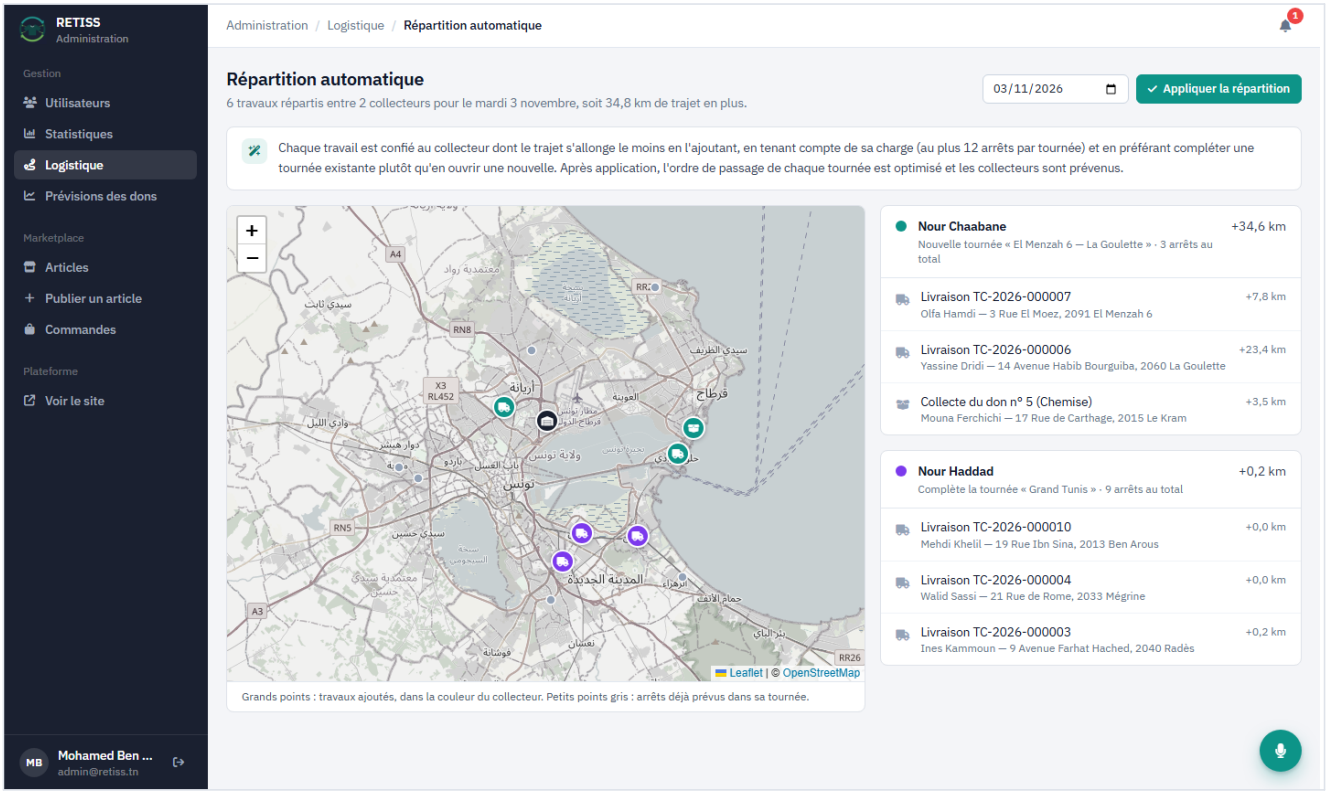
How to test

#	Do this	You should see
1	As collecteur@retiss.tn, open "Tunis Centre — Lac" and click **Mode tournée** (try it on a phone or with the browser's phone view, F12 → device toolbar).	The current stop (Montplaisir, "En cours") with Itinéraire, Terminé and the failure button.
2	Click **Terminé**, type a note, attach a photo, confirm.	The next stop is shown; the progress bar moves; the photo appears as a "Photo" link on the tournée page.
3	Click "Impossible de faire cet arrêt" and confirm without a reason.	"Indiquez la raison de l'échec." Pick "Client absent" and confirm: the stop is recorded as failed.
4	Allow location when the browser asks.	The chip says "Position partagée" and your blue dot appears on the map.
5	As admin, open Logistique → **Carte en direct** in another browser.	The collecteur's position and progress; within 10 seconds it reflects the changes made on the phone.
6	Close every stop of a tournée.	Tour mode shows "Tournée terminée"; the tournée status becomes Terminée.

Note. browsers only share location on HTTPS or on localhost. On a phone reaching your PC over plain HTTP the position will be refused; use the deployed HTTPS version or a tunnel for a real phone test.

5.4 Automatic assignment

Algorithm (parallel cheapest insertion). For every job waiting to be planned (paid orders to deliver, donations to collect) and every active collecteur, the engine computes how much the collecteur's route would grow if the job were inserted at its best position. It then adds a workload penalty (3 km per stop already planned) and a penalty for opening a new tournée (8 km), and assigns the cheapest pair, again and again, until everything is placed or every tournée is full (12 stops). The result: jobs on the way of an existing round join it almost for free, distant clusters go to different collecteurs. Jobs that cannot be placed are listed with the reason.



Assignment proposal for tomorrow: map coloured by collecteur, and per collecteur the jobs and the extra kilometres.

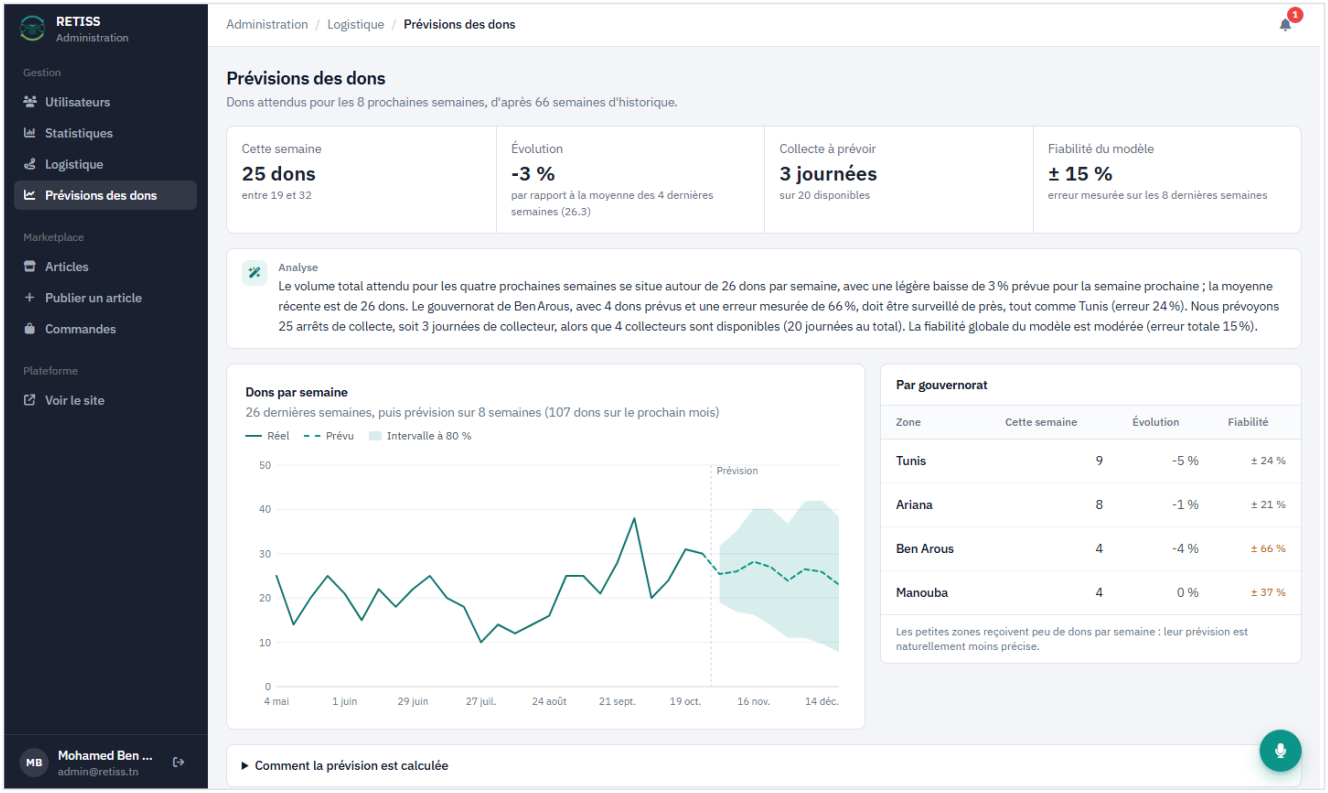
How to test

#	Do this	You should see
1	As admin, Logistique → **Répartir automatiquement** (in the "À planifier" card).	Tomorrow's proposal: the southern deliveries join Nour Haddad's "Grand Tunis" (+0.2 km); the north-east ones open a new tournée for Nour Chaabane.
2	Change the date at the top.	The proposal is recomputed for that day (never in the past).
3	Click **Appliquer la répartition**.	Tournées created or completed, each one already optimised; "À planifier" is empty; collecteurs get a bell notification "Nouvelles missions".
4	Log in as the collecteur who received jobs.	The bell shows the notification; the tournée lists the new stops in optimised order.

5.5 Donation-volume forecast (AI)

Model, kept simple so it can be explained: (1) donations are counted per week and per governorate (donor's address); (2) the platform's growth is removed with a linear regression, then each week of the year gets a seasonal coefficient learned on the past year and smoothed over 5 weeks, shared by all zones so a small noisy zone does not invent its own seasons; (3) level and trend are tracked with Holt's double exponential smoothing; (4) an 80 % range widens with the horizon, from the model's past errors; (5) **reliability is measured**: the model is replayed with the last 8 weeks hidden and compared with what really happened. On the demo data the total is accurate to about ±15 %.

The page also converts the forecast into resources (collection days needed vs available) and a language model writes a short briefing from these numbers only; it is forbidden to invent causes such as weather or holidays.



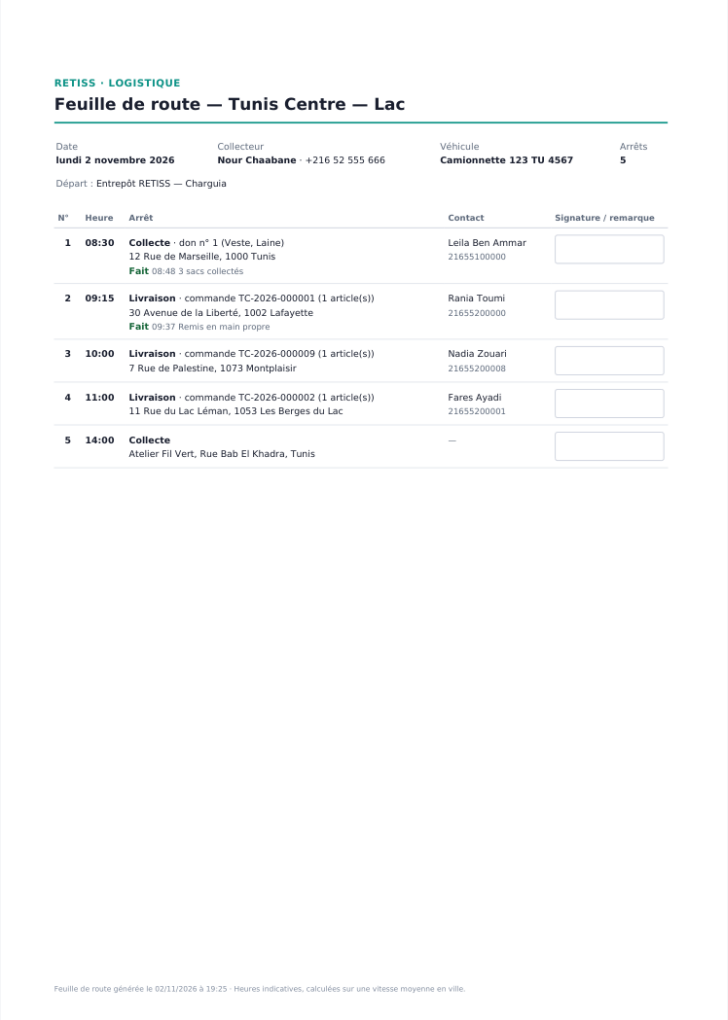
Forecast: key figures, AI briefing, history and 8-week forecast with its 80 % range, and per-governorate table.

How to test

#	Do this	You should see
1	As admin, click **Prévisions des dons** in the sidebar.	About 25 donations this week (range ~19–32), the trend, "3 journées sur 20 disponibles", "± 15 %".
2	Hover the chart.	A crosshair and a tooltip: real count for past weeks, forecast and range for future weeks.
3	Look at last year in the chart.	The August dip and the autumn peak that the seasonal coefficients learned.
4	Read the per-governorate table.	Small zones show a larger error, highlighted in amber when above 30 %.
5	Open "Comment la prévision est calculée".	A plain-language explanation of the method.

5.6 Route sheet, alerts and notifications

- **Feuille de route** (PDF): stops in order with times, what each one carries, contact and phone, done/failed marks with notes, and a signature box per stop. Available to the collecteur (own tournées) and to the admin (any tournée).
- **Failure alerts**: when a stop fails, every active admin receives a notification with the reason and a link to the tournée.
- **Notification bell**: the team's shared bell was added to the top bar of my admin and logistics pages, so these alerts and the "Nouvelles missions" notifications are visible there.



RETISS · LOGISTIQUE

Feuille de route — Tunis Centre — Lac

Date	Collecteur	Véhicule	Arrêts
lundi 2 novembre 2026	**Nour Chaabane** · +216 52 555 666	**Camionnette 123 TU 4567**	**5**

Départ : Entrepôt RETISS — Charguia

N°	Heure	Arrêt	Contact	Signature / remarque
1	08:30	**Collecte** · don n° 1 (Veste, Laine) 12 Rue de Marseille, 1000 Tunis **Fait** 08:48 3 sacs collectés	Leila Ben Ammar 21655100000	
2	09:15	**Livraison** · commande TC-2026-000001 (1 article(s)) 30 Avenue de la Liberté, 1002 Lafayette **Fait** 09:37 Remis en main propre	Rania Toumi 21655200000	
3	10:00	**Livraison** · commande TC-2026-000009 (1 article(s)) 7 Rue de Palestine, 1073 Montplaisir	Nadia Zouari 21655200008	
4	11:00	**Livraison** · commande TC-2026-000002 (1 article(s)) 11 Rue du Lac Léman, 1053 Les Berges du Lac	Fares Ayadi 21655200001	
5	14:00	**Collecte** Atelier Fil Vert, Rue Bab El Khadra, Tunis	—	

Feuille de route générée le 02/11/2026 à 19:25 · Heures indicatives, calculées sur une vitesse moyenne en ville.

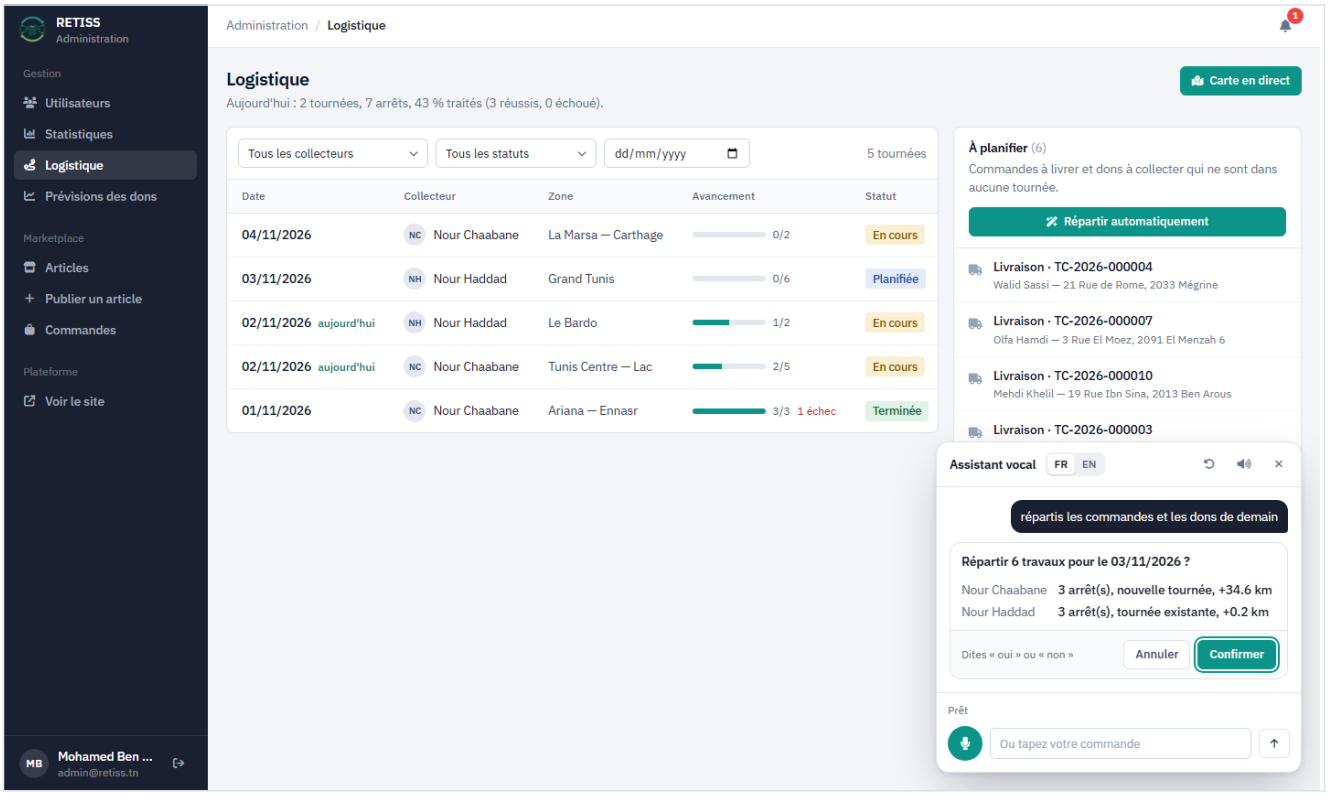
Printable route sheet (PDF).

How to test

#	Do this	You should see
1	On any tournée page, click **Feuille de route**.	A PDF named feuille-de-route-DATE-ZONE.pdf downloads.
2	As admin, open a tournée from Logistique and click **Feuille de route**.	Same PDF for any collecteur's tournée.
3	As a collecteur, mark a stop as failed; then log in as admin.	The bell counter increased; the alert opens the tournée.

6. AI voice assistant

A microphone button (bottom right, or Alt+V) on every admin and logistics page opens a conversation panel. You speak in French or English (FR/EN switch); the panel keeps its conversation from page to page.



The assistant proposing an action: nothing is written until you confirm (button, or say "oui" / "yes").

6.1 How it works

Step	What happens	Code
1. Listen	The browser records your voice and stops by itself after about 1.6 s of silence.	assistant.blade.php
2. Transcribe	Whisper (Groq) turns the recording into text, guided by the names of your users or zones. Noise or invented text is discarded.	Transcripteur.php
3. Understand	The language model receives the command, the list of allowed actions for your role, and the live data: users and statistics, today's tournées with positions and lateness, the forecast (admin) or your tournées and stops (collecteur).	Interpreteur.php, Catalogue.php, Contexte.php
4. Check	The proposed action is re-validated with the same rules and permissions as the forms. A failure goes back to the model once so it can fix it or ask.	Executeur.php
5. Confirm	Anything that writes data is shown as a card and waits for your confirmation; searches and navigation happen at once.	Executeur.php
6. Answer	Replies are read aloud: natural Orpheus voice in English, browser voice in French.	Synthetiseur.php

6.2 What you can say

Who	Example	Result
Admin	"Ajoute un utilisateur Karim Mansouri, e-mail karim arobase exemple point tn, rôle collecteur"	Card to confirm; a temporary password is generated and shown once.
Admin	"Désactive Kareem Mansoori et passe-le recycleur"	Finds Karim Mansouri despite the spelling; one card with both changes.
Admin	"Show me the active recyclers" / "ouvre les statistiques"	Opens the filtered list / the page.
Admin	"Fais-moi un résumé de l'activité des comptes"	Spoken analysis of the account statistics.
Admin	"Où en est Nour Chaabane ?"	Progress, next stop, lateness, distance from the nearest stop.
Admin	"How many donations next week?"	Forecast with its range.
Admin	"Répartis les commandes et les dons de demain"	Card with the split per collecteur; applied on confirmation.
Collecteur	"Ajoute une collecte à 9 h 30 au 12 rue de Marseille"	Uses today's tournée; card to confirm.
Collecteur	"Marque la mission 2 comme terminée" then "et la 3 a échoué, client absent"	Two updates; the second keeps the context.
Collecteur	"Optimise ma tournée de demain" / "lance le mode tournée"	Card with the gain / opens tour mode.
Collecteur	"Qu'est-ce qu'il me reste à faire aujourd'hui ?"	Spoken summary of the remaining stops.

How to test

#	Do this	You should see
1	Put your Groq key in .env (ASSISTANT_API_KEY), restart the server, log in as admin.	The teal microphone button at the bottom right.
2	Click it, choose FR, press the microphone and say one of the examples above (or type it).	Your sentence appears, then an answer or a confirmation card.
3	On a card, say "non" (or click Annuler); repeat and say "oui".	Nothing changes on "non"; the action runs on "oui" and the page opens.
4	Switch to EN and ask a question.	Everything (cards, errors, voice) is in English.
5	Navigate to another page.	The panel stays open with its conversation; the reset button (circular arrow) clears it.
6	As a collecteur, try "supprime l'utilisateur Karim".	Refused: the action does not exist for that role, whatever the AI proposes.

Note. the free Groq tier allows a few commands per minute. When the quota is reached the panel counts down and retries by itself.

7. Automated tests

Every feature has feature tests; AI replies and external services (Groq, OpenStreetMap) are simulated so the tests are fast, free and reproducible. Run them with:

```
php artisan test                                    (whole project: 423 passed, 2 skipped)
php artisan test tests/Feature/Logistique          (logistics only)
php artisan test --filter=OptimisationItineraireTest   (one file)
```

Test file	Tests	What it covers
Admin/UserManagementTest	13	Access control, search and filters, create/edit/delete, validation, self-protection rules, statistics.
Logistique/TourneeMissionTest	15	Tournée and mission CRUD, validation, ownership, stop order.
Logistique/LocalisationEtSuiviTest	16	Locating stops (order, donor, geocoding, cache), order/donation synchronisation, linking rules, admin overview.
Logistique/OptimisationItineraireTest	10	Zigzag reordering, optimality against all permutations, frozen and in-progress stops, times, preview/apply, stale previews, AI explanation.
Logistique/SuiviTempsReelTest	11	Tour mode, start/done/failed with photo and reason, automatic tournée status, GPS positions and throttling, live map data.
Logistique/AffectationAutomatiqueTest	7	Insertion on the way, clusters, capacity, unplaceable jobs, applying, notifications, stale proposals.
Logistique/PrevisionDonsTest	8	Flat, growing and seasonal histories, range, zones, current week excluded, page, AI briefing.
Logistique/FinitionsLogistiqueTest	5	Route sheet PDF and its access rules, failure alerts to admins, notification bell.
AssistantVocalTest	41	Every voice action and its confirmation, roles, languages, misheard names, multi-step commands, transcription, voice, logistics commands.
Total for my module	**126**	

8. Code map

Area	Files
Admin dashboard	app/Http/Controllers/Admin/UserController.php, StatistiqueController.php · app/Services/StatistiquesUtilisateurs.php · app/Http/Middleware/EnsureAdmin.php · resources/views/admin/layouts, users, partials, statistiques.blade.php
Logistique CRUD	app/Models/Tournee.php, Mission.php · app/Http/Controllers/Logistique/TourneeController.php, MissionController.php · app/Http/Middleware/EnsureCollecteur.php · resources/views/logistique/…
Locations and sync	app/Observers/MissionObserver.php · app/Services/Logistique/Geocodeur.php, LocalisationMission.php, SynchronisationMission.php
Optimisation	app/Services/Logistique/OptimiseurItineraire.php, ExplicationItineraire.php · Logistique/OptimisationController.php
Tracking and maps	app/Models/PositionCollecteur.php · Logistique/ConduiteController.php · Admin/LogistiqueController.php · resources/views/logistique/partials/carte.blade.php (shared Leaflet helper)
Assignment	app/Services/Logistique/AffectationAutomatique.php · Admin/AffectationController.php · Notifications/MissionsAffecteesNotification.php

Area	Files
Forecast	app/Services/Logistique/PrevisionDons.php, BriefingPrevision.php · Admin/PrevisionController.php · database/seeders/HistoriqueDonsSeeder.php
Finishing	Logistique/FeuilleDeRouteController.php · Notifications/MissionEchoueeNotification.php
Voice assistant	app/Services/Assistant/* · app/Http/Controllers/AssistantController.php · resources/views/admin/partials/assistant.blade.php · config/assistant.php
Settings and data	config/logistique.php · database/migrations/2026_10_05_* and 2026_11_02_* · database/seeders/LogistiqueSeeder.php

9. Limits to keep in mind

- Distances and times are **estimates** (straight line × 1.3, 25 km/h, 10 min per stop), not real road routing.
- The forecast is only as good as the history; on the demo data it is generated, deterministic history. The 80 % range is a range, not a promise.
- Phone location needs HTTPS (or localhost). Real GPS and camera were exercised by automated tests and desktop checks, not on a physical phone.
- AI texts depend on the free Groq quota and on an API key; without them, computed summaries are shown instead and every page keeps working.
- The language model can misunderstand a sentence; this is why every write action is re-validated by the server and confirmed by the user.
- The shared seeders add about 1,250 past donations to the database, which also shows in other modules' donation statistics.